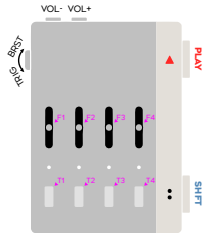


Wakes v0.8.0 - Pg.1 - Welcome



- Hey! I'm **Adara**. Welcome to **Wakes**, a custom firmware for **Teenage Engineering's** unreleased "SP-1" stem player. Wakes adapts code from **three** Eurorack modules into one device: **Plaits**, a macro-synthesizer, **Marbles**, a playful random sequencer, and **Yarns**, a signal converter. The modules were created by **Émilie Gillet**, otherwise known as **Mutable Instruments**.

- First and foremost, you should know how to turn the device **ON** or **OFF**. Holding down **SHFT** for 1.5 seconds will show a brief animation before the device will turn **ON**, going to its first page, **Plaits**.

- While **ON**, holding **SHFT** for 3 seconds will turn the device **OFF**.

- While plugged in, Wakes has a **STANDBY** mode. The side LED's show the battery state.

- While **ON**, any input while **SHFT** is held down will interrupt any ongoing shutdown behaviours, allowing you to work inside **SHFT** pages.

- Even if interrupted, holding down **SHFT** for 25 seconds will cause the face LEDs to flash quickly, notifying you that the SP-1 is about to perform an **emergency shutdown** in 5 seconds if you don't let **SHFT** go. Release and return to holding **SHFT** to get back to what you were doing. This is intended as an emergency backstop.

- At any point, even while **OFF**, tapping **SHFT** for a short moment will bring up a battery meter.

- While using **Wakes**, you'll be bouncing between **Plaits** to design sounds, and **Marbles** for sequencing and modulation.

- As you bounce back and forth, the face LED's will show you what's going on through a system of **glyphs**.

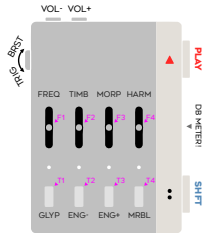
- Faders **F1-F4** have a catch-up mechanism with the backend values, so moving them and swapping pages/engines doesn't create unexpected behaviours. For clarity, **follow the LED's** to know what the true parameters are.

- Each module has a **SHFT** page containing advanced parameters and options that open up avenues for experimenting.

- Double-tapping **SHFT** enters a deep tinkering page called **SETTINGS**. There are two **SETTINGS** pages: One for **Plaits**, one for **Marbles**.

- Legend: ○ - Lit LED ● - Unlit LED





- **Wakes'** first page is **Plaits**: A synthesizer containing a multitude of sound engines representing different sound design techniques, with parameters attached to explorative macros. A **macro-synthesizer**.

- **Plaits** has four main faders: **FREQ (F1)**, **TIMB (F2)**, **MORP (F3)**, and **HARM (F4)**. Each of these controls several components of the selected synthesis engine.

- Plaits has a **low-pass gate**, controlled by an internally-routed **decay envelope**.

- For now, you can read about what the main parameters do for each engine in **commonly available Plaits guides**.

- While in Plaits, the four side LED's will show a dB meter. Each LED represents 6dB of dynamic range.

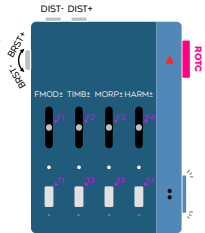
- Generally speaking, **FREQ** will control the overall pitch of the sound. **TIMB** sweeps the spectral content from dark/sparse to bright/dense. **MORP** explores lateral timbral variations. **HARM** controls the frequency spread/overall balance of the spectrum.

- The rocker switch plays notes, triggering the lowpass gate. Press it down to send a single **TRIG**, pull it up to send a **BRST**!

- Engines are known by **glyphs**. Pressing **GLYP (T1)** will flash the glyph representing the selected engine. Hit **ENG+ (T2)** or **ENG- (T3)** to browse engines.

- **PLAY** engages **Marbles'** clock, which is patched by default to trigger Plaits and give it musical notes.

- Press **MRBL (T4)** to go to **Marbles**.



- Hold **SHFT** to go to **Plaits SHFT**:

- Each fader here (**F1-F4**) represents a **modulation depth control** for incoming control signals. By default, Plaits' **decay envelope** is assigned to all of the four parameters. Use the faders to modulate Plaits' macros! Fun!

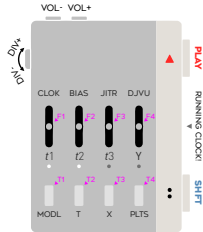
- **BRST-** and **BRST+** select **BRST**'s rhythmic subdivision: **1/1** through **1/128**.

- **DIST-** and **DIST+** control the depth of a **master softclipper**. Loud and crunchy!

- Holding down any of **T1-T4** for 2 seconds **Unpatches** Marbles modulators from that parameter.

- Holding down **PLAY** for 3 seconds **Rips Out Plaits' Cables**, returning it to its default configuration!

Wakes v0.8.0 - Pg.3 - Marbles - t Side (left)

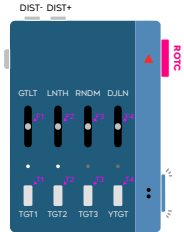


- **Marbles** has two sides: **t** and **X**. **Glyphs** show which side you are on: ○○○● for **t** and ●○○○ for **X**. Each side has **3** outputs: **t1-t3**, **X1-X3**. An extra output **Y** is offered.
- Moving **F1-F4** on either page shows the fader values briefly, before returning to showing the individual **t** and **X** outputs.
- **t** is a random gate generator. It emulates a player following a stable metronome.
- **CLOK** (**F1**) controls BPM (default: 120).
- **t2** is **always** a stable clock: 50% on, 50% off. **BIAS** (**F2**) controls the distribution of gates coming out of **t1** and **t3**.
- **JTR** (**F3**) controls how chaotically gates land on ticks. Accurate, to super drunk.
- **DJVU** (**F4**) controls the probability of repeating patterns. At 50% it's a locked loop. Down from 50% adds random notes. Up from 50% reorders memorized notes.

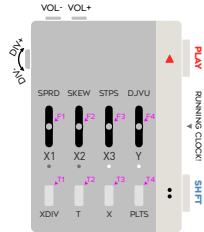
- **MODL** (**T1**) selects **BIAS**' distribution model for **t1** and **t3** activity:
 - cointoss: gates distribute evenly.
 - clusters: random pairs of clock subdivisions appear at each side.
 - drums: **BIAS** steps through drum pattern tables containing rhythms.
 - indpcoin: each side tosses its own coin. **t1** and **t3** can coincide.
 - divider: **BIAS** selects from fixed subdivision pairs. **8:1 > 1:1 > 1:8**.
 - threestates: a three-way choice per clock tick: **t1**, **t3** or neither.
- **DIV-/DIV+** multiply the clock: $\frac{1}{4}$, **1**, **4**.
- Pressing **T2** shows you the current page.
- Press **X** (**T3**) to go to the **X** side of Marbles, or **PLTS** (**T4**) to go back to **Plaits**.

- Hold **SHFT** to go to **t SHFT**:
- The four buttons (**T1-T4**) select **Plaits modulation targets** for **t1-t3**, or for **Y**. The mod targets are (in order): ●○○○-**TRIG**, ●●○○-**LEVL**, ○●●●-**FM**, ●○○●-**TIMB**, ●●○○-**MORP**, ●●○○-**HARM**, ●●○○-**none**. For **Y**, ●○○○ routes to **FREQ**.
- **GTLT** (**F1**) scales **t1** and **t3**'s max outputs.
- **LNTH** (**F2**) determines the length of on-states for **t1** and **t3** gate signals.
- **RNDM** (**F3**) determines gate length randomization for **t1** and **t3**.
- **DJLN** (**F4**) sets the maximum **DJVU** loop length: **1-8**, **10**, **12**, **14** and **16** beats.
- Holding down any of **T1-T4** for 2 seconds **Unpatches** any Plaits destinations.
- Hold **PLAY** for 3 seconds to **Rip Out Marbles' Cables**, returning it to its default state.

Marbles - t SHFT



Wakes v0.8.0 - Pg.4 - Marbles - X Side (right)



- **X** is a random modulation generator. Its three outputs, **X1-X3**, are clocked by the outputs of Marbles' **t** side, and generate random control signals that can vary from **Sample & Hold** style steps to **smooth, meandering modulations**. It can modulate Plaits' parameters, or be quantized to play musical intervals, creating melodic sequences, as well as enabling many other ideas.

- **Y** shows up on this page as well. It's a smooth, meandering full-range random source that's clocked at **1/8th** the rate of **t2** by default. **Y** never obeys **DJVU**, she's wild like that.

- **SPRD** (**F1**) spreads the concentration of signals around the central point.
- **SKEW** (**F2**) skews the average signal intensity up or down.

- **STPS** (**F3**) smooths or tightens the vertical steppiness of the control signals. Below-centre, the signals have random intensities. Above-centre, it **quantizes X** to a musical scale.

- **DJVU** (**F4**) is the same as on the **t** side. Both sections **t** and **X** are controlled by the same memory.

- Press **XDIV** (**T1**) to choose the intensity diversity among the outputs: **identical, bump** (**X2** is strongest) or **tilt** (**X1** is weakest, **X3** is strongest).

- Press **t** (**T2**) to go back to **t**.

- Press **PLTS** (**T4**) to go back to **Plaits**.

Hold **SHIFT** to go to **X SHIFT**:

- The buttons [**T1-T4**] select **Plaits modulation targets** for **X1-X3**, or for **Y**. The targets are **FM**, **TIMB**, **MORP**, **HARM**, **FREQ**, **LEVL**, **none**. **FREQ** is the default input for musical tunings, whilst **FM** is full-range for experimentation.

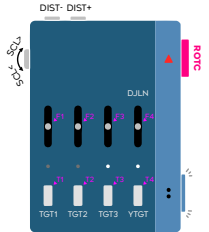
- **DJLN** (**F4**), like in **t**, controls the length of the **DJVU** sequence. **DJVU** and **DJLN** are the same across the two sides.

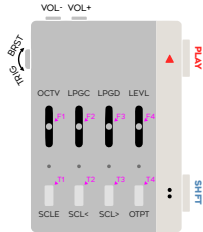
- **SCL** select from a range of scales to **quantize** into, when **STPS** is above its midpoint. View default available scales in the Marbles **SETTINGS** guide page.

- Holding down any of **T1-T4** for 2 seconds **Unpatches** any Plaits destinations.

- Hold **PLAY** to **Rip Out Marbles' Cables**.

Marbles - X **SHIFT**





- Reach Plaits' **SETTINGS** page by double-tapping **SHFT** while in the Plaits module. Single-tapping **SHFT** will return you to where you were.

- **OCTV's** (**F1**) default is its maximum value. When not maxed, it locks Plaits' available frequencies to ± 7 semitones around a chosen octave. Choose the range to play by moving the fader. Maximize it to go full-range again. Its minimum value is also full-range, but without the C4 centre-detent, for the freaks out there.

- **LPGC** (**F2**) lets you control the **color** of the **low-pass gate**, which is composed of a **low-pass filter** and an **amplifier**. Lower values mix in more of the low-pass filter, making the **internal decay envelope** and **LEVL** sound duller in tone. Higher values use more of the amplifier, making the sound brighter.

- **LPGD** (**F3**) lets you change the length of the **decay envelope**. Higher values set longer lengths, making triggered notes take longer to decay. Any self-patching from the **SHFT** attenuverters will also be longer. Shorter values sound snappier and quieter overall.

- **LEVL** (**F4**) offsets **Plaits'** filter and amplifier. Moving it fades the sound in and out. The color of this fade is affected by **LPGC**. Setting **LEVL** to maximum will effectively **bypass the low-pass gate**, allowing **Plaits** to **drone**. Some Plaits engines will react beautifully to modulating **LEVL**.

- **Wakes** includes a **smart Plaits quantizer** feature. It lets you noodle around with notes on a scale by **moving FREQ**, without Marbles' modulation. It's set to ******-off** by default. Activate it by pressing **SCL>** (**T2**). Deactivate it by pressing **SCL<** (**T3**) until the glyphs show ******-off** again. Once activated, it first disables any Marbles routings going into **FREQ**. Then, it takes over **FREQ** and quantizes it. Combine it with **OCTV** to lock your available notes to a short range. **SCLE** (**T1**) flashes the currently-active **quantizer scale's glyph**. Scales are the same as Marbles', refer to the Marbles **SETTINGS** manual page for their glyphs.

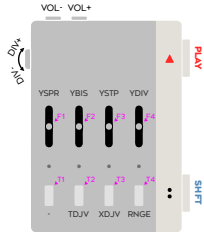
- **Plaits** features an extra output, **AUX**, containing **additional content from each engine**. **Wakes** gives you **OTPT** (**T4**) on this page to select from different output configurations: ******-OUT**, ******-AUX**, ******-OUT+AUX**, and the **Wakes-unique ****-OUTxAUX**.

Plaits Tips

- Plaits is designed to be really breakable. Experiment with fast **BRST**, quick Marbles modulations, lots of mod targets. **Just go crazy with it**, and see what wilderness you end up in. **Ripping Out The Cables** will take you back home.

- **FM** is a legacy Plaits input containing exp-TZFM for Plaits. It's used for atonal modulations.

- You can **TRIG** while **LEVL** is open ;)



- Reach Marbles' **SETTINGS** page by double-tapping **SHIFT** while in the Marbles module. Single-tapping **SHIFT** will return you to where you were.

- **YSPR (F1)** lets you control Y's spread, same way you do for X.

- **YBIS (F2)** lets you control Y's randomization skew towards higher or lower signal strengths.

- **YSTP (F3)** lets you control its overall steppiness. Lower values are more sinusoidal. Central values are linear, triangular. Higher values are slewed squares, up to an even sample & hold.

- **YDIV (F4)** lets you choose at which subdivision of the clock Y is clocked. This lets you accelerate it to higher values if you want.

- **TDJV (T2)** and **XDJV (T3)** toggle **DJYU** for the f or X sides on or off. **DJYU** is on for both by default. If you toggle it off for one of the sides, that side won't be affected by **DJYU**'s looping algorithms.

- **RNGE (T4)** sets the control signal strength range for the X and Y outputs. The options are: ○●●● - 0-20%, ○○○● - 0-50%, ○○○○ - ±50%, ●●●● - **INTL**. **INTL** is the default - a **Wakes**-unique upside of combining the two modules in a closed circuit: It decides your signal strength range selection for you, per-output, automatically, and with the target Plaits engine in mind. That said, if you still want to play around with unexpected results, the other modes remain available.

Scales:

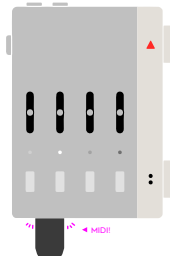
- There are 6 default scales Wakes offers for quantizing the X outputs or **Plaits'** **FREQ (F1)** fader. When you press **SCL<** or **SCL>** on Marbles' **SHIFT** page, their **glyphs** will flash on the face LEDs as you cycle through the list.

- Available Scales: ○●●●-major, ○●●●-minor, ○○○●-pentatonic, ○○○○-pelog, ●○○○-bhairav, ●●○○-shri.

- If you choose one of these scales, then move **STPS (X F3)** above-center, then send one of the X outputs into **FREQ**, Marbles will tell Plaits to play notes on that scale.

Marbles Tips

- You can combine multiple Marbles outputs into one Plaits destination. Using **Unpatch** disconnects only that specific output from its destination.



- Wakes supports **MIDI IN and Audio OUT** over the SP-1's **USB-C** port. Connect your SP-1 to a **USB host**, and it should pop up as a **MIDI** and **USB audio** device.

- Wakes automatically routes pitch and MIDI notes into **Plaits**, and routes MIDI clock into **Marbles**.

- By default, Wakes will decide control ranges and scale your CC appropriately for unipolar and bipolar parameters across the device.

- **Wakes' repo includes config files: config/midi.ini and config/audio.ini. Review the files for advanced information.**

- **Fork the repo** to build your own configurations, and customize Wakes' USB handling to your needs.

PLAITS - MIDI BEHAVIOURS

- Plaits will now receive **pitch, pitchbend (± 2 default), gate, velocity** (scales **LEVL**) and **MIDI CC** information from incoming MIDI signals. This bypasses the quantizer.

- **FREQ's** centre-detent will now expand to 10% making it easier to lock onto correct MIDI pitch. Moving **FREQ** will now transpose Plaits.

- Plaits is a **monophonic** voice. It does not support polyphony outside of 3 engines: **Chords, String** and **String Machine**, and those have distinct polyphony behaviours that don't constitute traditional polyphony.

- **Plaits' MODEL** input is now exposed via MIDI CC, letting you select engines from the host. Caution - high CPU.

MARBLES - MIDI BEHAVIOURS

- **Marbles** will sync to external MIDI clocks until MIDI is disconnected. MIDI START is now the same as pressing **PLAY**.

- Marbles' **CLOK** (**t F1**) fader now **selects a ratio** of the incoming MIDI clock which Marbles plays on. The ratios are: 1/4, 1/3, 1/2, 2/3, 1 (at centre), 3/2, 2, 3 or 4.

- Marbles' **DIV+** and **DIV-** rocker switch controls multiplying ranges of that subdivision: $\circ\bullet\bullet\bullet$ -1/4, $\bullet\circ\circ\bullet$ -1, and $\bullet\bullet\bullet\circ$ -4.

- Wakes compares incoming MIDI note timing with Marbles' clock, and runs Marbles that much ahead, so Marbles' steps land with your notes and your DAW needs no clock offset (leave it at 0).

DEFAULT MIDI CC LIST

Plaits

3-TIMB, 9-MORP, 14-HARM, 15-FREQ, 20-FMOD $\pm$, 21-TIMB $\pm$, 22-MORP $\pm$, 23-HARM $\pm$, 24-LPGC, 25-LPGD, 26-LEVL, 102-OCTV, 105-MODEL, 64 - Sustain pedal.

Marbles

t

27-CLOK, 16-BIAS, 85-JITR, 17-LNTH, 86-RNDM,

X

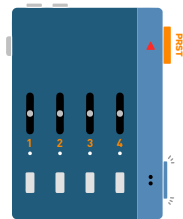
28-SPRD, 29-SKEW, 30-STPS,

Y

18-YSPR, 19-YBIS, 87-YSTP, 104-YDIV

DJVV

31-DJVV, 103-DJLN



STORAGE

- **v0.8.0** introduces **device storage**, which allows users to store and retrieve files for later features.
- **First-time users:** The SP-1 stores samples in a 4GB eMMC chip. Wakes needs to format the SP-1's storage to operate. This will clear any data in that partition if you haven't backed it up (you probably don't need to care about what's there by default). When you first go to flash Wakes, you have a choice: **wakes-sp1.bin** or **wakes-sp1-fresh.bin**. If you've never flashed the **fresh** one before, use **fresh** - It'll format your partition once per flash. After flashing **fresh** once, you can then flash **wakes-sp1.bin** in future versions to preserve your onboard storage. Newer versions might break **PRSTs**, in which case you should flash **fresh** again.

- Wakes will behave like a removable mass-storage card reader while in its **STANDBY** state (**OFF** and USB is plugged in). While storage activity is ongoing, you'll see a dancing pattern on the face LED's. To protect files, you can't turn the SP-1 **ON** while storage activity is ongoing. If you want to unplug your SP-1 from your host, wait until the lights are fully dark before doing so, otherwise you risk file corruption. If the host is just a MIDI device, don't worry about unplugging abruptly; Wakes won't let it touch the storage.
- When Wakes is turned **ON**, it disconnects its storage from the host. To manipulate stored files, Wakes must be turned **OFF**. If you've ejected Wakes' storage while plugged in, you'll need to replug the SP-1 to your host for the drive to reappear.

PRST - SAVESTATES

- **v0.8.0** also introduces **PRST**, allowing you to save and retrieve **full Wakes patches** that you've worked on as **.PRST** files.
- Saved **PRSTs** contain **every Wakes setting and DJVU memory**, except for **VOL**, in the .ini format.
- When turned **ON**, Wakes always loads into the last-loaded **PRST** slot. It loads all slots from storage into RAM on boot. Effectively, while using Wakes, you are always operating within one of the slots.
- Wakes saves its state to the **currently selected PRST slot only** once turned **OFF** or to **STANDBY**. Moving to the next slot will **not** save Wakes' current state.

PRST - BROWSER

- While holding **SHFT**, tap **PLAY** once (instead of holding it to **ROTC**) to briefly enter the **PRST** browser, showing which slot you are currently on.
- While the animation is still playing, Tap **PLAY** again quickly to flip to the next slot.
- Letting go of **SHFT** or waiting a short moment takes you out of the **PRST** browser.